Lesson 10 Proofs in Number Theory

In Lessons 6 to 9, we have had a comprehensive discussion of basic proof techniques using elementary examples. In this lesson, we apply the same techniques to prove some basic theorems in number theory (the study of integers), focusing on **divisibility** and **greatest common divisor** (Lesson 1 Exercises).

Theorem 1 Let $a, b \in \mathbb{Z}$ and I be the set of positive linear combinations,

$$I = \{ax + by : x, y \in \mathbb{Z},\ \ ax + by > 0\}$$

The least (smallest) element of I is $\gcd(a, b)$.

Theorem 1 is an extremely important theorem in number theory, it will be proved in the Appendix. We give an example. The positive divisors of 8 and 20 are $1, 2, 4, 8$ and $1, 2, 4, 5, 10, 20$, so $\gcd(8, 20) = 4$. Theorem 1 says that the *least* positive linear combination of 8 and 20 is 4, which we may obtain as follows:

$$8(-2) + 20(1) = 4$$

Theorem 1 also says that there are *no* integers x and y such that $8x + 20y = 2$.

Recall that a and b are ***relatively prime*** if $\gcd(a, b) = 1$ (Lesson 1 Exercises #10).

Theorem 2 Integers a and b are relatively prime if and only if $ax + by = 1$ for some $x, y \in \mathbb{Z}$.

Proof If a and b are relatively prime, then by Theorem 1, there exist $x, y \in \mathbb{Z}$ such that $ax + by = \gcd(a, b) = 1$. Conversely, if $ax + by = 1$ for some $x, y \in \mathbb{Z}$, then 1, being the smallest positive integer, must be the smallest positive linear combination of a and b, and by Theorem 1, $\gcd(a, b) = 1$.

Theorem 2 is an extremely useful characterization of relatively prime numbers and is used to prove many theorems.

Theorem 3 Suppose $a, b, c \in \mathbb{Z}$. If a and b are relatively prime and $a|bc$, then $a|c$.

Proof If $\gcd(a, b) = 1$, then by Theorem 2, $ax + by = 1$ for some $x, y \in \mathbb{Z}$. If $\boldsymbol{a|bc}$, then $bc = ak$, for some $k \in \mathbb{Z}$, and

$$c = (ax + by)c = axc + y\boldsymbol{bc} = axc + n\boldsymbol{ak} = a(xc + yk)$$

Since $xc + yk$ is an integer, we have $a|c$.

Theorem 4 For all integers a, b and any prime p, if $p|ab$, then $p|a$ or $p|b$.

Proof If $p|a$, then we are done. If $p \nmid a$, then since the only positive divisors of p are 1 and p, the only common divisor of a and p is 1, and $\gcd(a, p) = 1$. By Theorem 3, $p|b$.

We have used Theorem 4 on a few occasions to prove other propositions, now we have finally proved Theorem 4 itself!

Appendix

Proof of Theorem 1

I is nonempty, as it has $a(a) + b(b) = a^2 + b^2 > 0$.

By WOP (Lesson 8 Exercises #11), I has a least element $d = ax + by$.

We show that $d|a$: otherwise by DA, $a = qd + r,\ 0 < r < d$, and we have an element of I which is smaller than d:

$$r = a - qd = a - q(ax + by) = a(1 - qx) - bqy$$

contradicting that d is a least element. Similarly, $d|b$. So, d is a common divisor of a and b. Let e be an arbitrary common divisor of a and b, then $e|a$ and $a|ax$ imply that $e|ax$ (Exercise 6 of Lesson 6), similarly, $e|by$. By Exercise 7 of Lesson 6, $e|(ax + by)$ for all integers x and y, so $e|d$, i.e. $d \geq e$, and $d = \gcd(a, b)$.

Notice that in the above proof, we also proved the following.

Proposition 5 Let $d = \gcd(a, b)$. Then $e|d$ for any common divisor e of a and b.